

AWS in an Online Food Delivery Platform

Today, online food delivery applications such as Zomato and Swiggy serve millions of customers every day. People can order food from different restaurants using their mobile phones, and restaurants receive orders instantly. Managing such a large application requires powerful, secure, and scalable cloud services. AWS (Amazon Web Services) provides many services that help these applications run smoothly and efficiently.

1. Amazon EC2 (Elastic Compute Cloud)

One of the most important AWS services used is **Amazon EC2 (Elastic Compute Cloud)**. EC2 provides virtual servers where the food delivery application is hosted. Whenever thousands of users open the application during lunch or dinner time, the traffic increases significantly. AWS can automatically increase the number of EC2 instances to handle the extra load. When the traffic becomes normal, it reduces the number of servers, helping the company save money because it pays only for the resources it uses.

2. AWS Database services

The application stores a large amount of information such as customer details, restaurant information, food menus, delivery partner information, payment records, and order history. This information is stored using **AWS Database services**. The database provides fast access to data and ensures that customers can quickly place orders without delays. It also helps restaurants update menus and prices in real time.

3. Amazon S3 (Simple Storage Service)

The platform contains thousands of food images, restaurant logos, invoices, and bills. Instead of storing these files on the application server, they can be stored in **Amazon S3 (Simple Storage Service)**. S3 is highly durable, secure, and can store a massive amount of data. Whenever a customer opens the application, food images are loaded from S3 quickly, providing a better user experience. S3 also helps reduce the load on the application servers.

4. AWS Lambda

AWS Lambda is another useful service because it allows the application to run small pieces of code without managing servers. For example, when a customer places an order, Lambda can automatically send an order confirmation email or SMS, update the restaurant about the new

order, generate an invoice, or notify the delivery partner. Since Lambda runs only when an event occurs, the company pays only for the execution time, making it cost-effective.

5. AWS Security and Privacy

Security is extremely important because the application stores sensitive information such as customer names, addresses, phone numbers, and payment details. **AWS Security and Privacy** services help protect this information. AWS Identity and Access Management (IAM) allows administrators to give different permissions to employees based on their roles. Data encryption protects customer information from unauthorized access, and AWS continuously monitors the system to detect security threats.

AWS also helps improve the reliability of the application. If one server fails, another server can immediately take over so that customers continue using the application without interruption. This high availability is very important because even a few minutes of downtime can affect thousands of orders and reduce customer trust.

6. AWS IAM (Identity and Access Management)

Security is one of the most important parts of any online application. AWS IAM (Identity and Access Management) controls who can access AWS resources. Different users receive different permissions based on their responsibilities.

For example:

- Customers can only view and manage their own accounts and orders.
- Restaurant owners can update only their own menu and orders.
- Delivery partners can view only the orders assigned to them.
- Developers can manage the application but cannot access customer payment details unless permission is granted.
- AWS administrators have full access to manage AWS services.

This role-based access control improves security and prevents unauthorized access.

7. Monitoring, Backup, and Reliability

AWS continuously monitors the health of servers and applications. If CPU usage becomes high or a server experiences problems, AWS can generate alerts so that the technical team can respond quickly. AWS also provides backup and disaster recovery services. If data is accidentally deleted or a server fails, backups can restore the information with minimal downtime. This ensures that customers can continue placing orders without major interruptions.